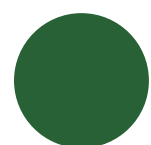




# AIOT LIDAR SAFETY SYSTEM (ALSS)

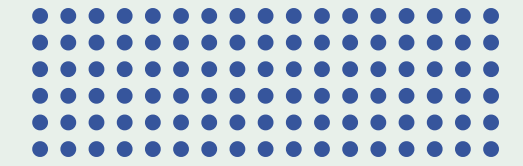


**The Coffee Team**  
**VNUHCM - University of Science**

START PRESENTATION →

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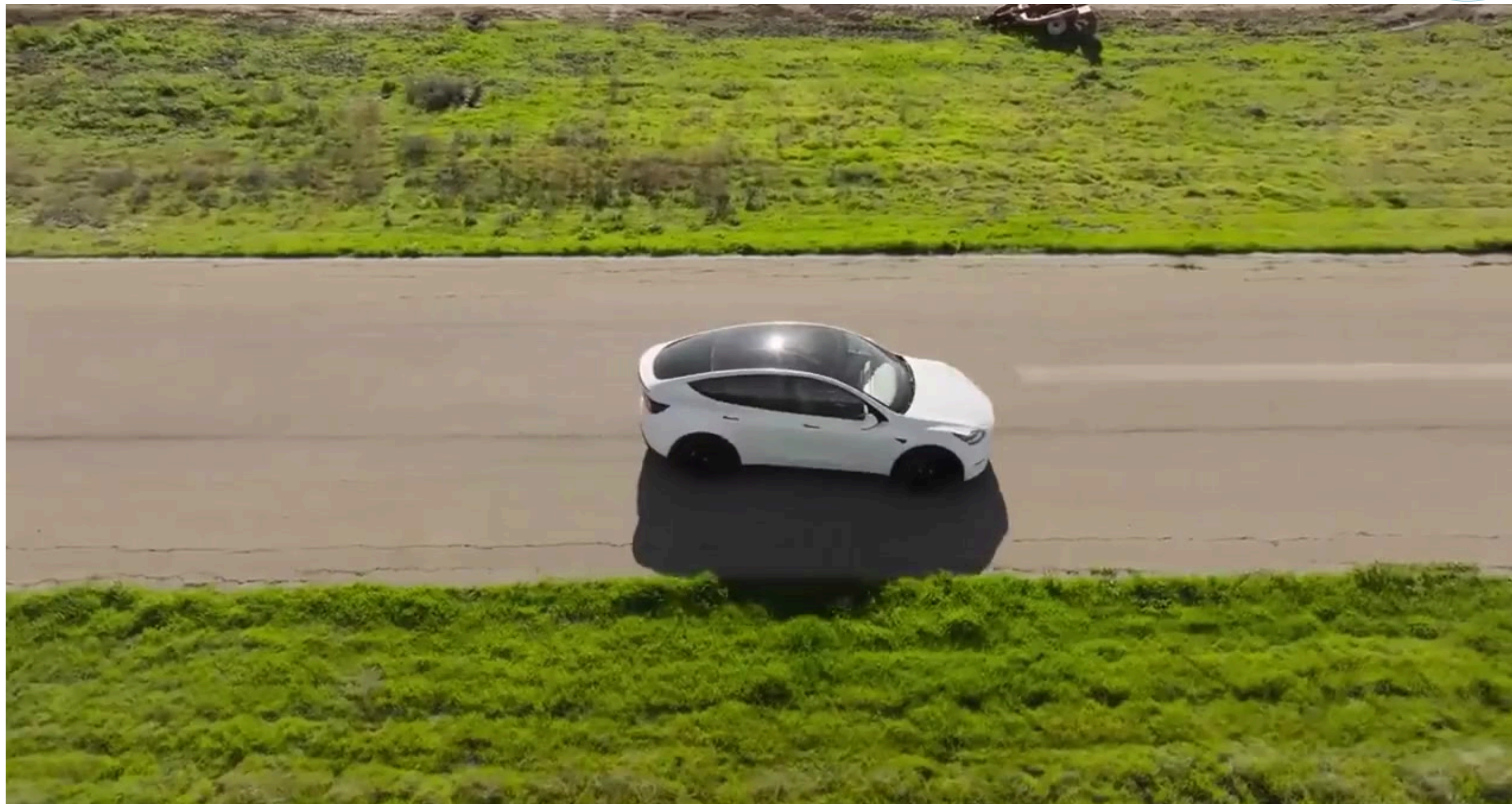


# 01 PROBLEM STATEMENT

- The problem that the System seeks to solve
- Real-world examples
- The pain points, inefficiencies associated with the problem.

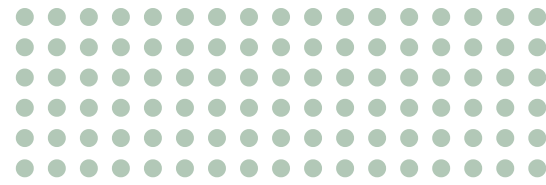


# When the camera fails



**Video source:** <https://youtu.be/IQJL3htsDyQ?si=M7lwL4ZNvMFSMmAd>





# Real-world examples

## Vehicle blind spot

**A car was sideswiped when it entered the blind spot of a truck.**

<https://vnexpress.net/xe-con-bi-ui-ngang-khi-roi-vao-diem-mu-cua-xe-tai-4808432.html>

**Lane Change Leads to Car Crash in Truck's Blind Spot**

<https://vnexpress.net/xe-con-bi-dam-xoay-ngang-vi-ruc-nach-xe-tai-4851514.html>



## Railway crossing

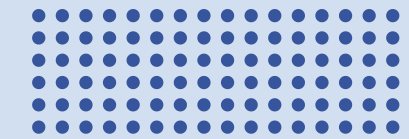
**Two killed as train hits pickup truck in Dong Nai Province.**

<https://e.vnexpress.net/news/news/traffic/2-dead-as-train-crashes-into-pickup-truck-in-southern-dong-nai-province-4775104.html>

**Train crashes into taxi in central Vietnam, killing two.**

<https://e.vnexpress.net/news/news/train-crashes-into-taxi-in-central-vietnam-killing-two-3950248.html>





# The problem that the ALSS seeks to solve

## Complex Traffic Conditions in Vietnam



Vietnam's traffic is dense and complex, with diverse vehicles and high accident risks due to poor visibility and human errors.

## High-Risk Zones



Traditional traffic safety measures (e.g., static signs, manual monitoring) are ineffective in dynamically preventing collisions in real-time.

## Limits of Traditional Safety Measures



Static signs and manual supervision are ineffective for real-time collision prevention.

## Key Focus Areas



To address these challenges, the system targets three high-risk scenarios: **factory gate access/exist**, **blind spots of large vehicles**, and **railway crossings**.

# The pain points, inefficiencies associated with the problem.

Limited visibility

Slow Reaction Time

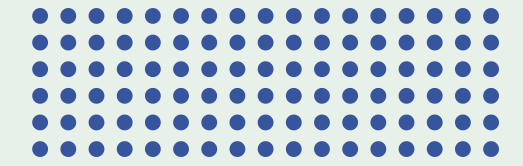
Inefficient Traffic Control

Manual Monitoring Inefficiency

Lack of Collision Prediction & Prevention:

Lack of AI & IoT Integration





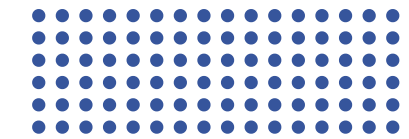
# 02

# SOLUTION

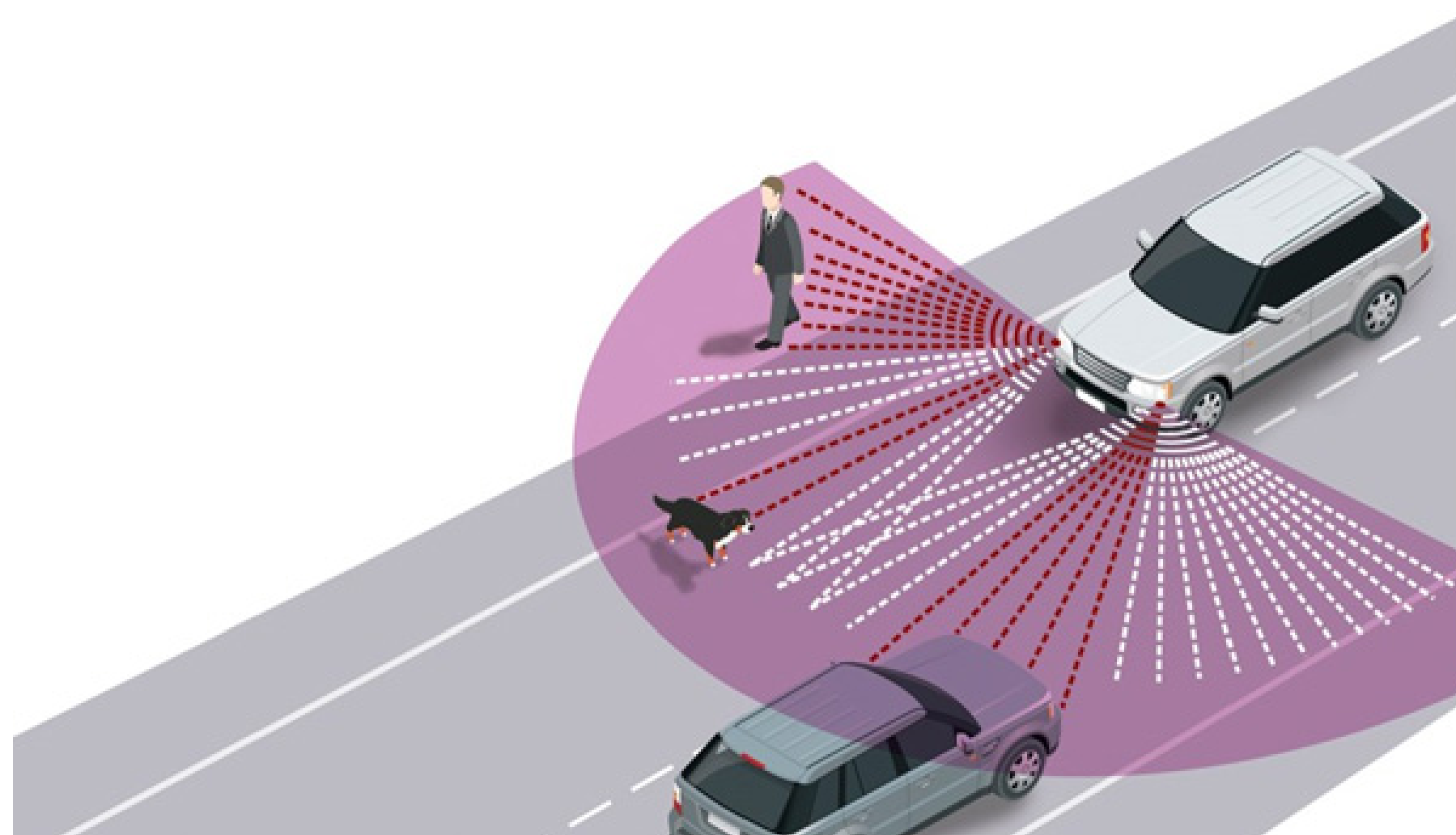
# OVERVIEW

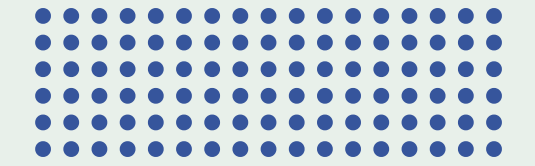


# Solution Overview



To address the identified challenges, this solution proposes developing an **AIoT-powered system** that integrates **real-time LiDAR monitoring** to enhance vehicle safety.





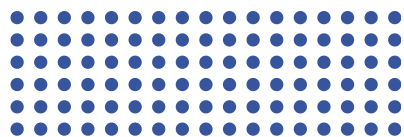
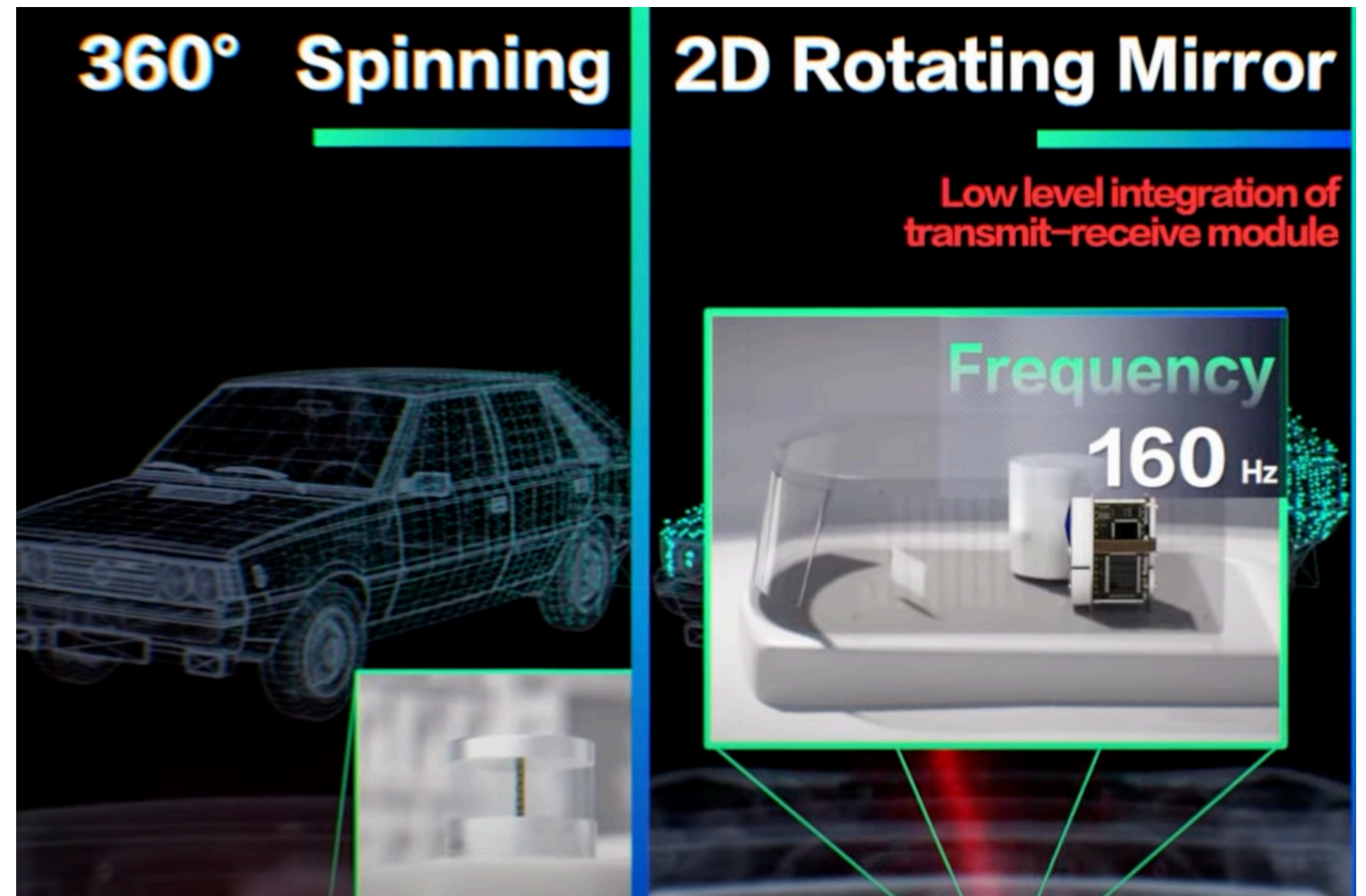
# 03 CORE FUNCTIONALITY

- **Real-Time Detection**
- **AI-Driven Collision Prediction**
- **Immediate Alert System**
- **AIoT Integration for Smart Safety Measures**
- **Target Users**
- **Responsible AI**



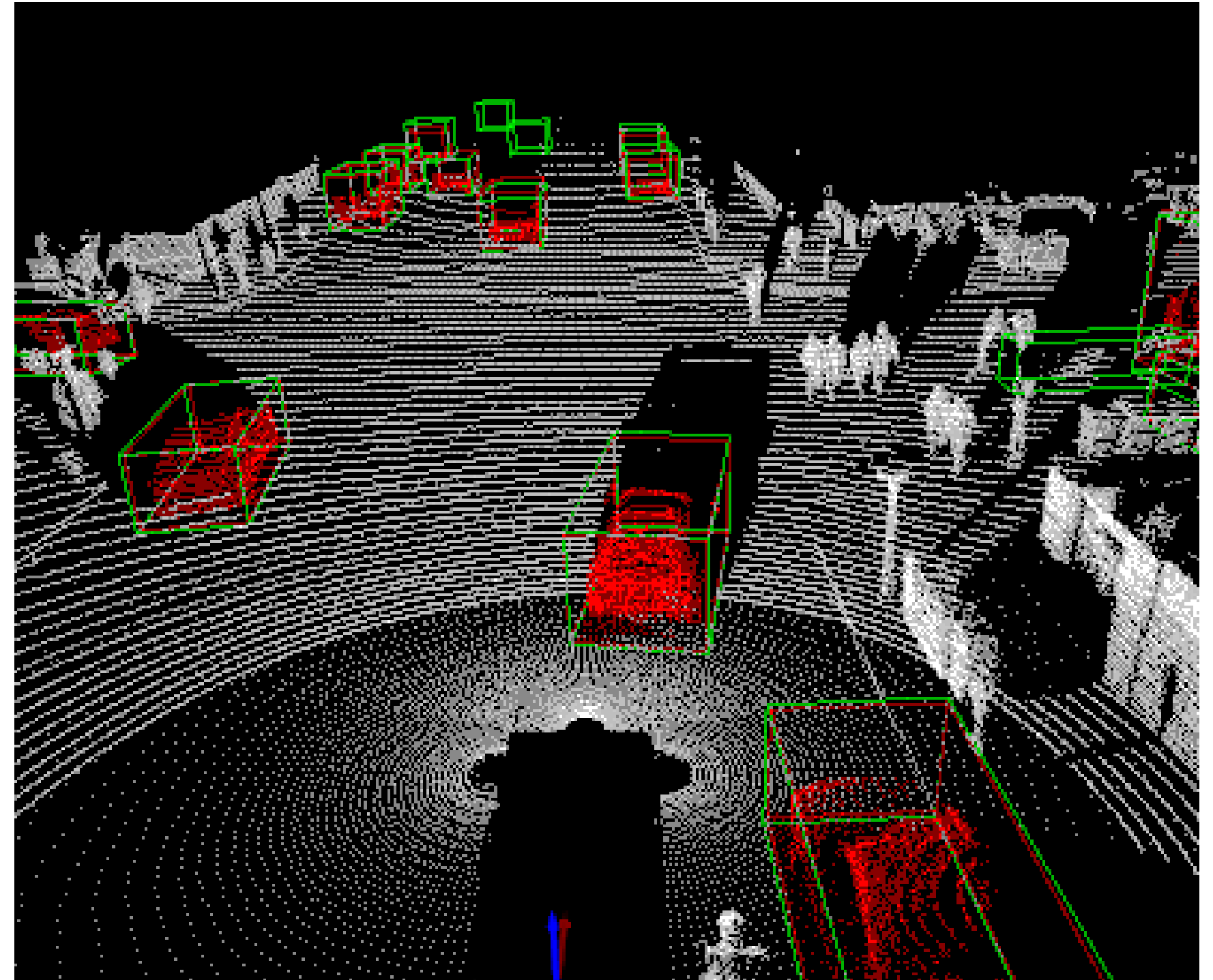
Uses 360° Mechanical Spinning LiDAR or 2D Rotating Mirror LiDAR to continuously scan surroundings.

# Real-Time Detection



# AI-Driven Collision Prediction

Deep learning models (CNN & LSTM Hybrid) analyze real-time data to detect potential accidents.



# Immediate Alert System

Real-time warnings for drivers via visual and audio alerts .

Helps prevent accidents by providing early notifications for critical situations.





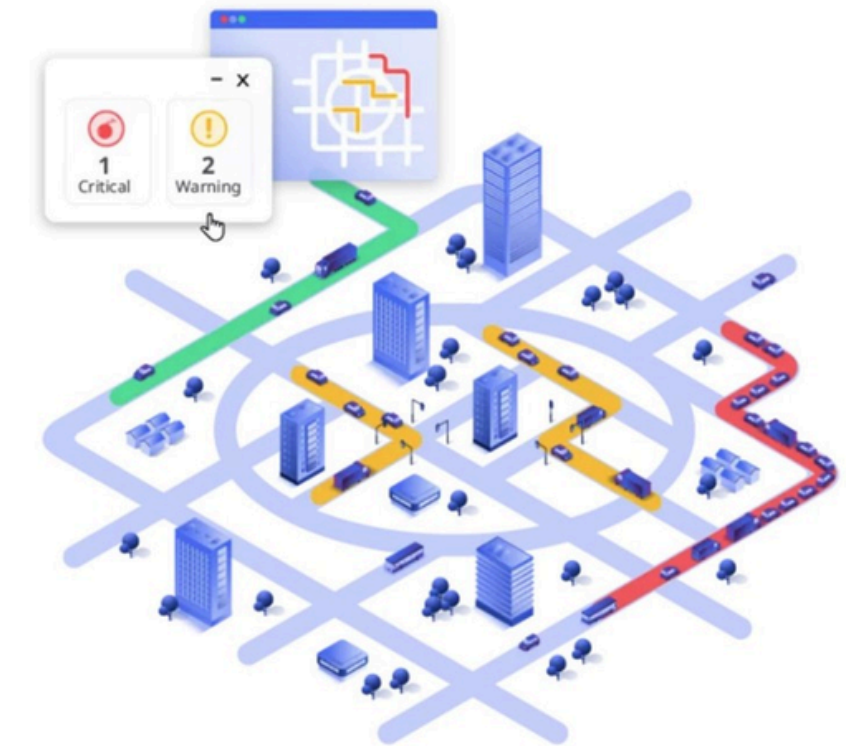
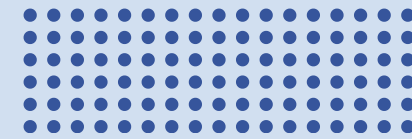
# AIoT Integration for Smart Safety Measures

Connects with vehicle safety systems such as brakes, airbags, and display panels.

Automatically applies safety measures if the driver fails to react in time.



# Target Users



## Personal vehicle users

Real-time collision alerts to improve situational awareness.

## Commercial vehicle owners

Asset protection, accident cost reduction.

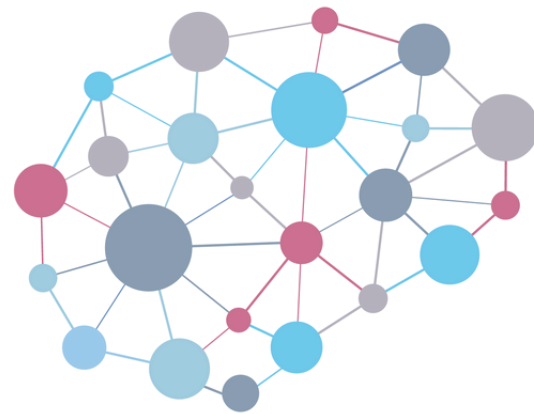
## Urban Traffic Management

Data collection for safer traffic planning.





# Responsible AI



To improve the limitations for vehicle drivers while traveling, we propose implementing a system that combines IoT and AI to detect, analyze, predict, and prevent collisions in potentially dangerous traffic situations.

## Fairness



Trained on diverse traffic scenarios to ensure equitable safety measures for all road users (drivers, pedestrians, cyclists)

## Privacy and Security



Strong encryption and cybersecurity measures prevent data breaches.

## Reliability and Safety



The system undergoes rigorous testing in multiple traffic environments to ensure accuracy.

## Inclusiveness



Supports multilingual and accessible interfaces for different user groups.

## Transparency



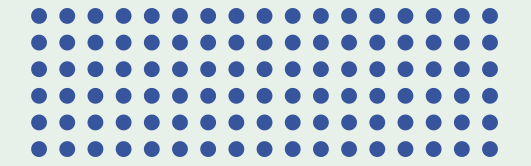
Traffic authorities and users can access detailed reports and AI decision logs to verify accuracy.

## Accountability



AI-driven decisions are logged and traceable, ensuring accountability in case of errors or failures





# 04

# ARCHITECTURE

- **Architecture**
- **Dataset**
- **AI Model And Testing**
- **System powered by AIoT**

# Architecture

## LiDAR Dataset

nuScene v1.0-trainval  
Google Cloud Storage/ Drive for  
data storage

## Collision Prediction AI Model

Google Colab Pro as the training  
environment  
Hybrid CNN  
Long Short-Term Memory  
(LSTM)

## User Interface & Visualization

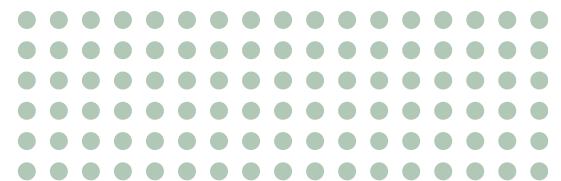
Open3d, PyGame  
PyQt5  
Matplotlib

## Object Detection

1x spinning LIDAR  
(Velodyne HDL32E)

## Testing

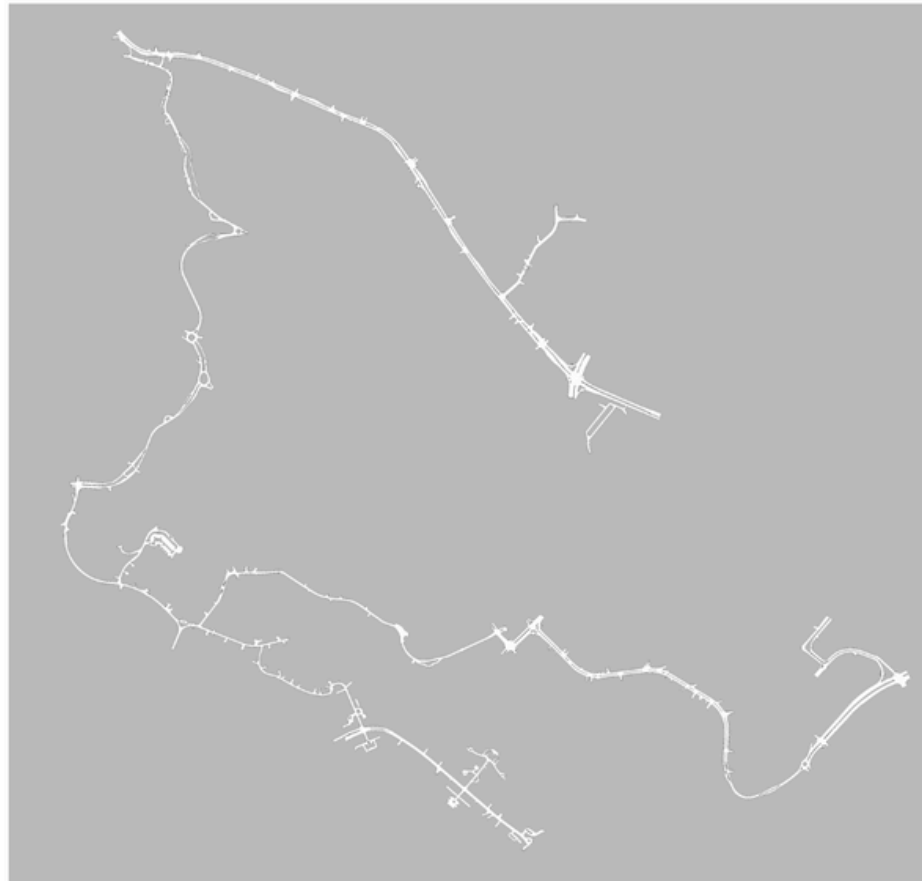
nuScenes v0.1-teaser version  
nuScenes v1.0-test version



# Dataset - Scene planning



Boston Seaport



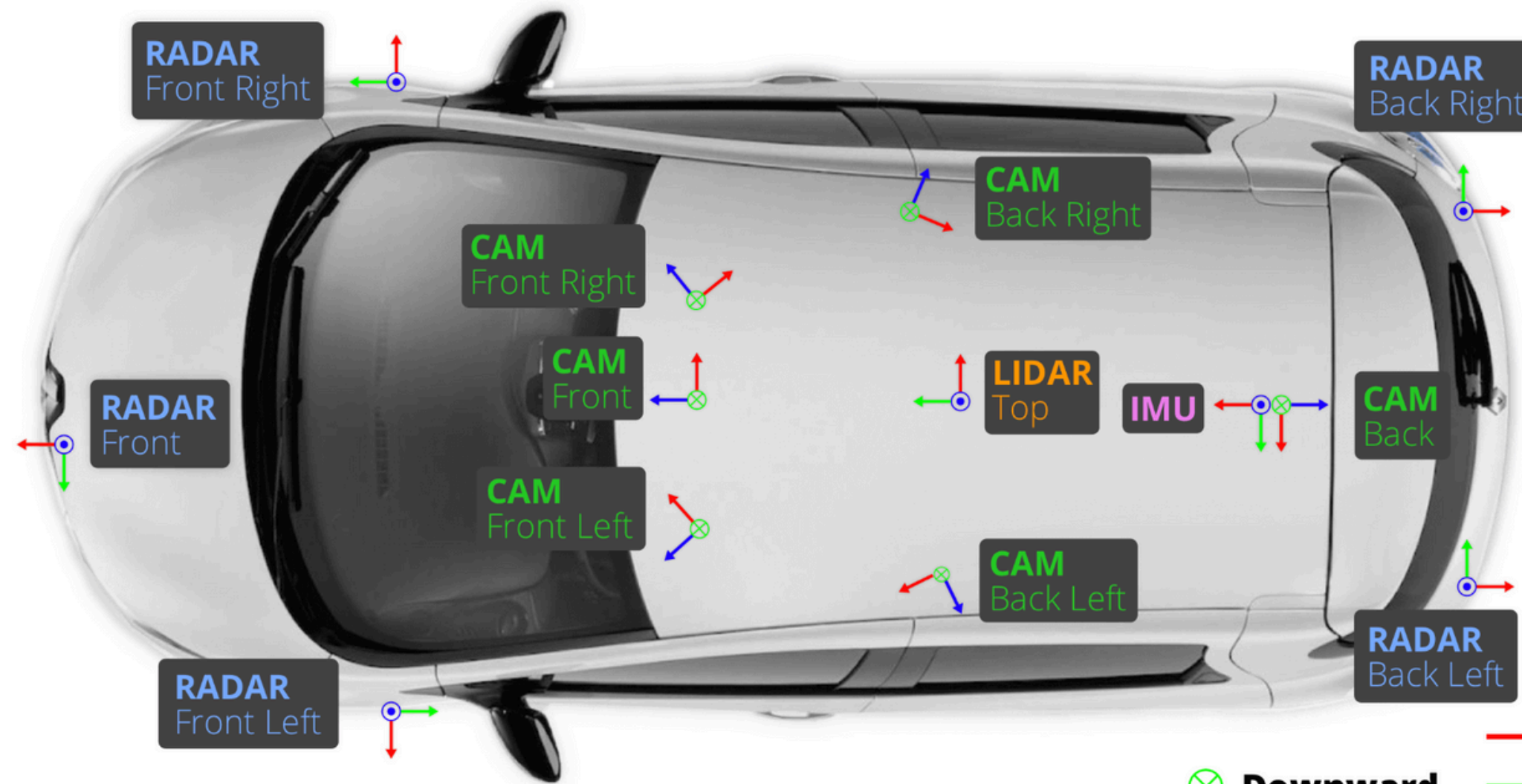
Singapore Queenstown

“For the nuScenes dataset we collect approximately 15h of driving data in Boston and Singapore...Driving routes are carefully chosen to capture challenging scenarios. We aim for a diverse set of locations, times and weather conditions. To balance the class frequency distribution, we include more scenes with rare classes (such as bicycles). Using these criteria, we manually select 1000 scenes of 20s duration each. These scenes are carefully annotated using human experts.”



# Dataset - Car Setup

Using nuScenes Open Dataset. The AI model just uses the LiDAR data which getting from Velodyne HDL32E installed at the top of the car (see the Car Setup image).



1x spinning LIDAR (Velodyne HDL32E):

20Hz capture frequency

32 beams, 1080 (+-10) points per ring

32 channels

360° Horizontal FOV, +10° to -30° Vertical FOV, uniform azimuth angles

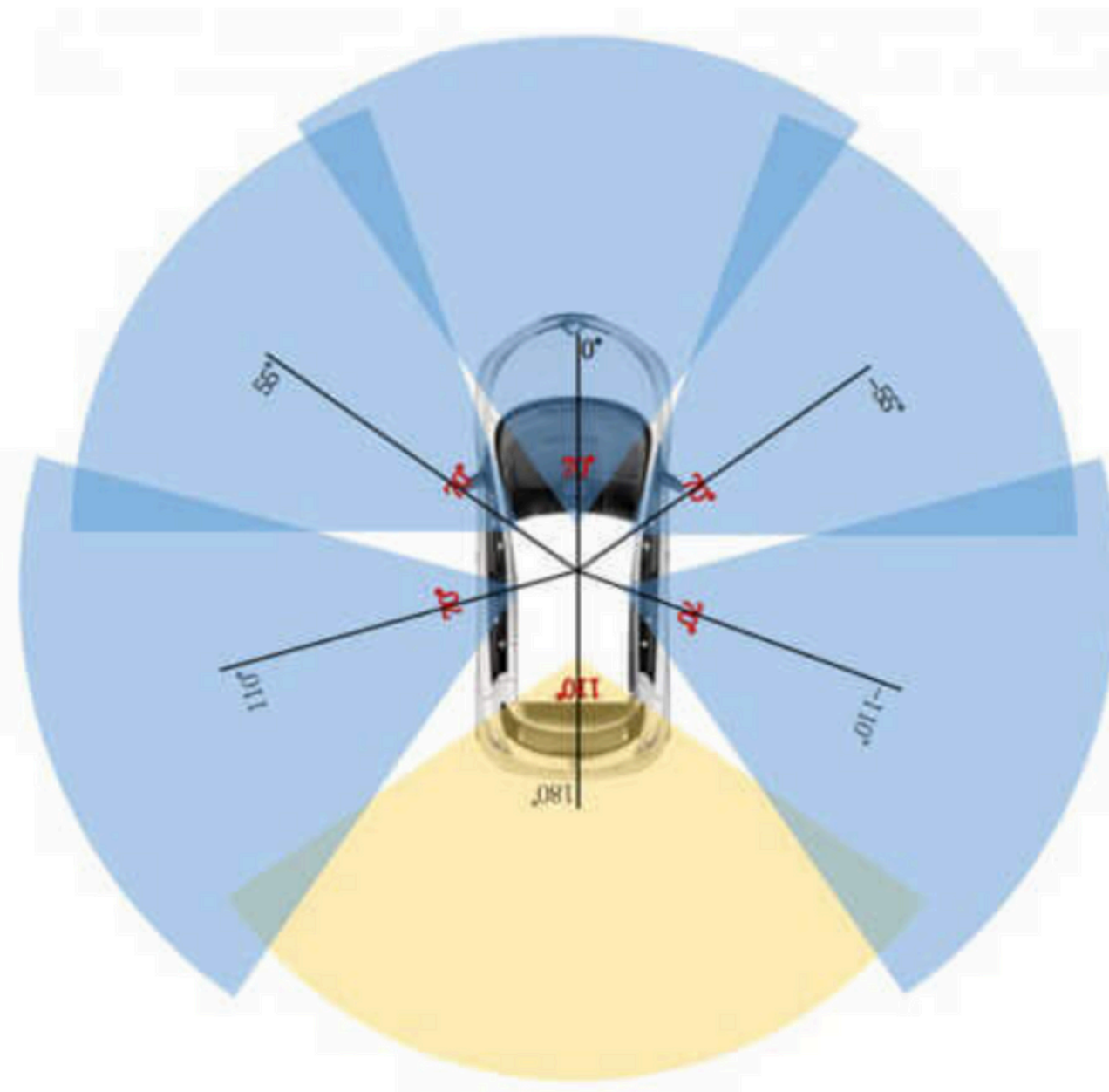
80m-100m Range, Usable returns up to 70 meters,  $\pm 2$  cm accuracy

Up to ~1.39 Million Points per Second



# Dataset - Sensor Calibration

To ensure high-quality multi-sensor data, all sensors must be calibrated in terms of extrinsics and intrinsics. Extrinsics are defined relative to the ego frame (center of the rear axle).

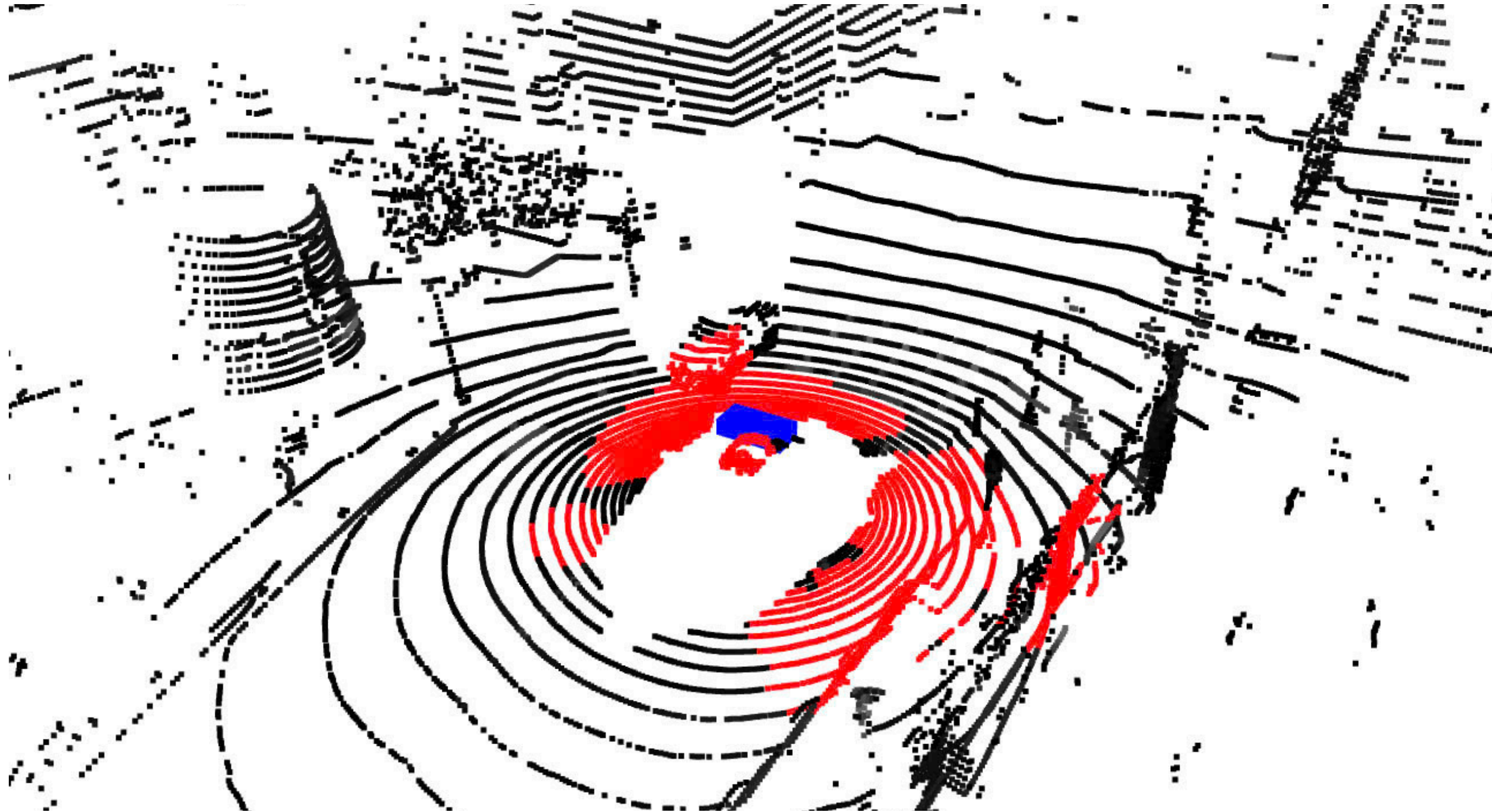


## LIDAR extrinsics:

We use a laser liner to accurately measure the relative location of the LIDAR to the ego frame.



# Dataset - Illustration

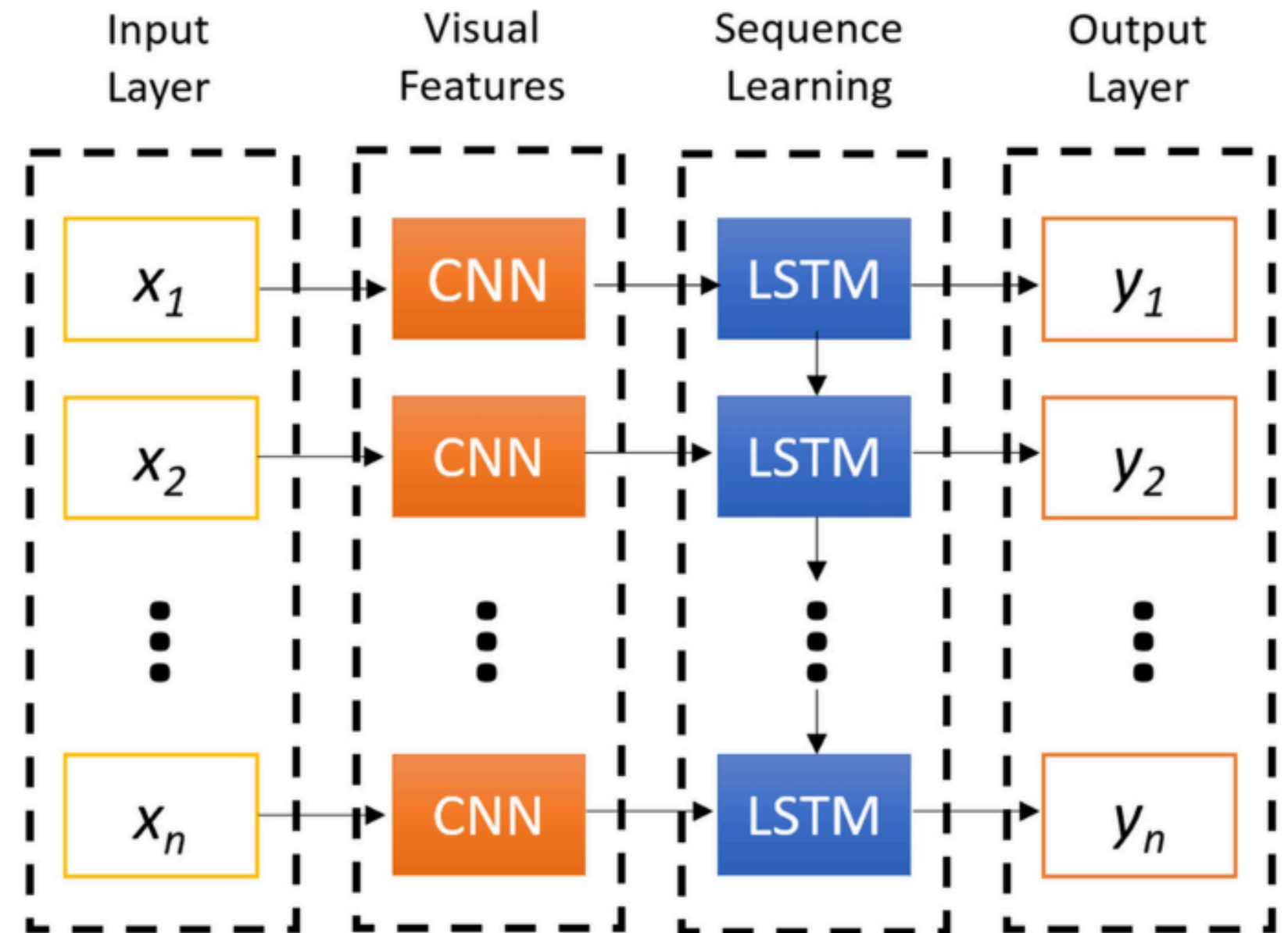


*Source: as part of our project implementation*



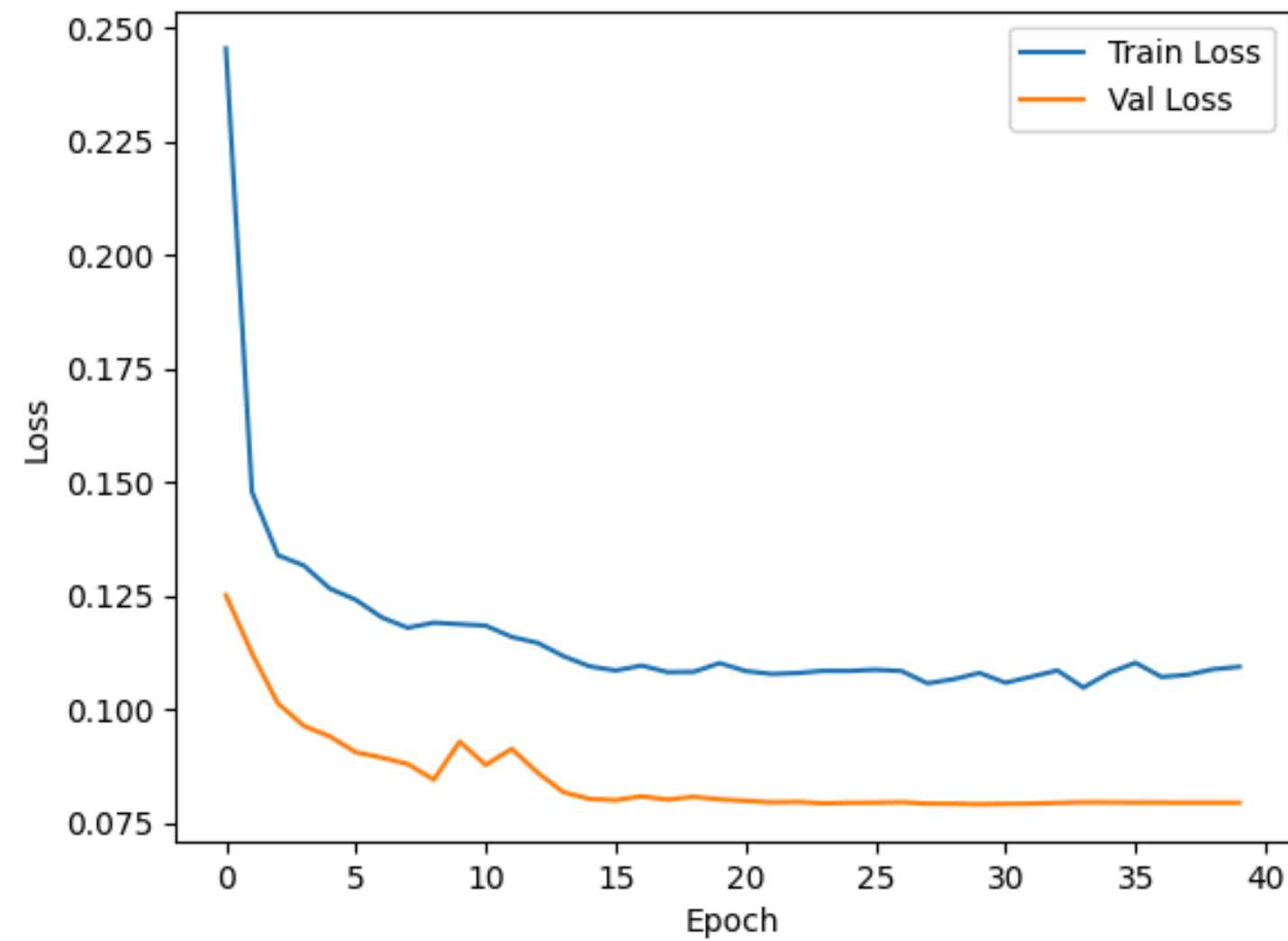
# Motion Prediction Model

Trained on Google Colab, the Hybrid CNN-LSTM model leverages CNNs to extract spatial features from vehicle states (position, velocity, heading, relative distance) and LSTMs to model temporal dependencies, making it ideal for predicting future velocities from sequential LiDAR data.

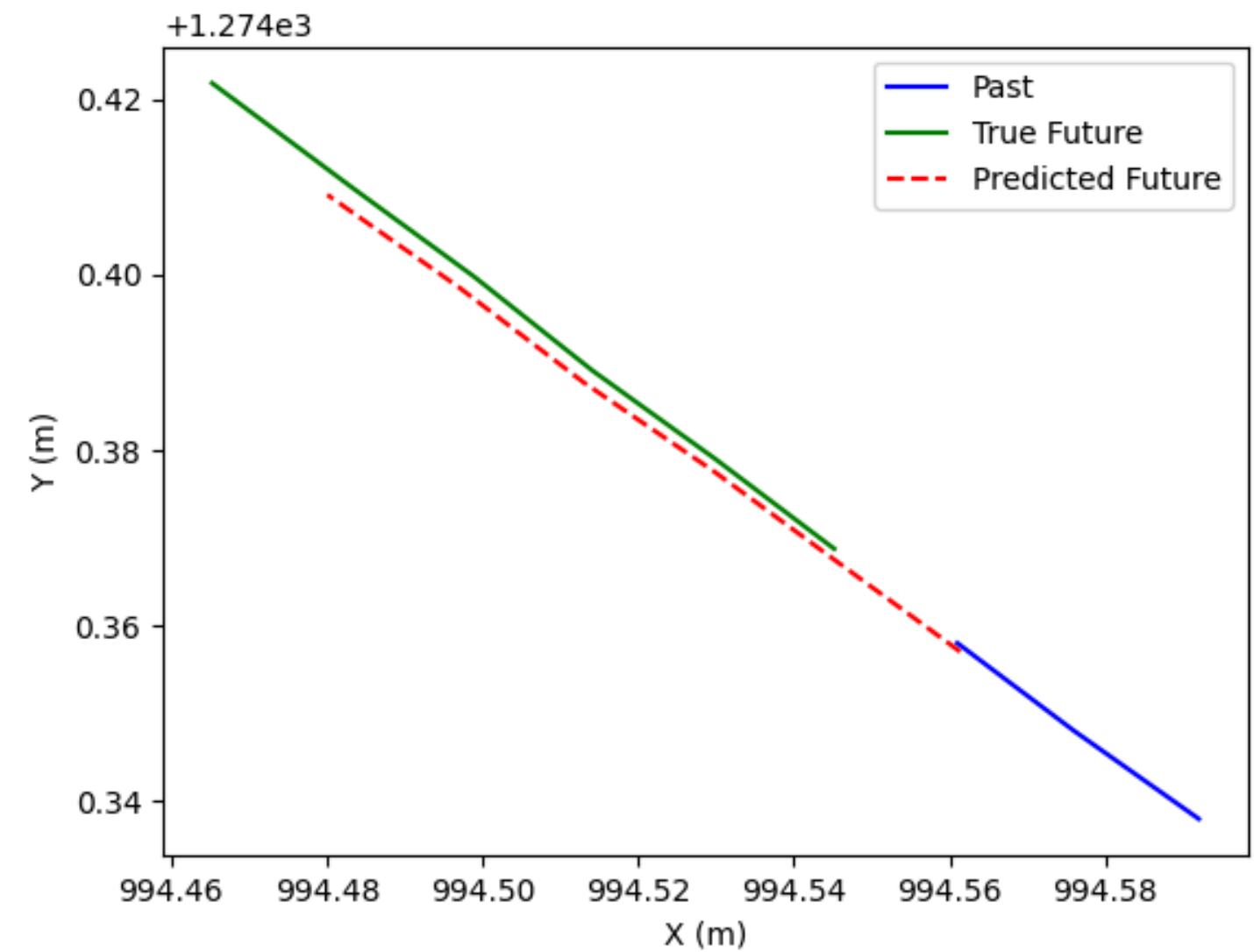


Hybrid CNN-LSTM

# Motion Prediction Model

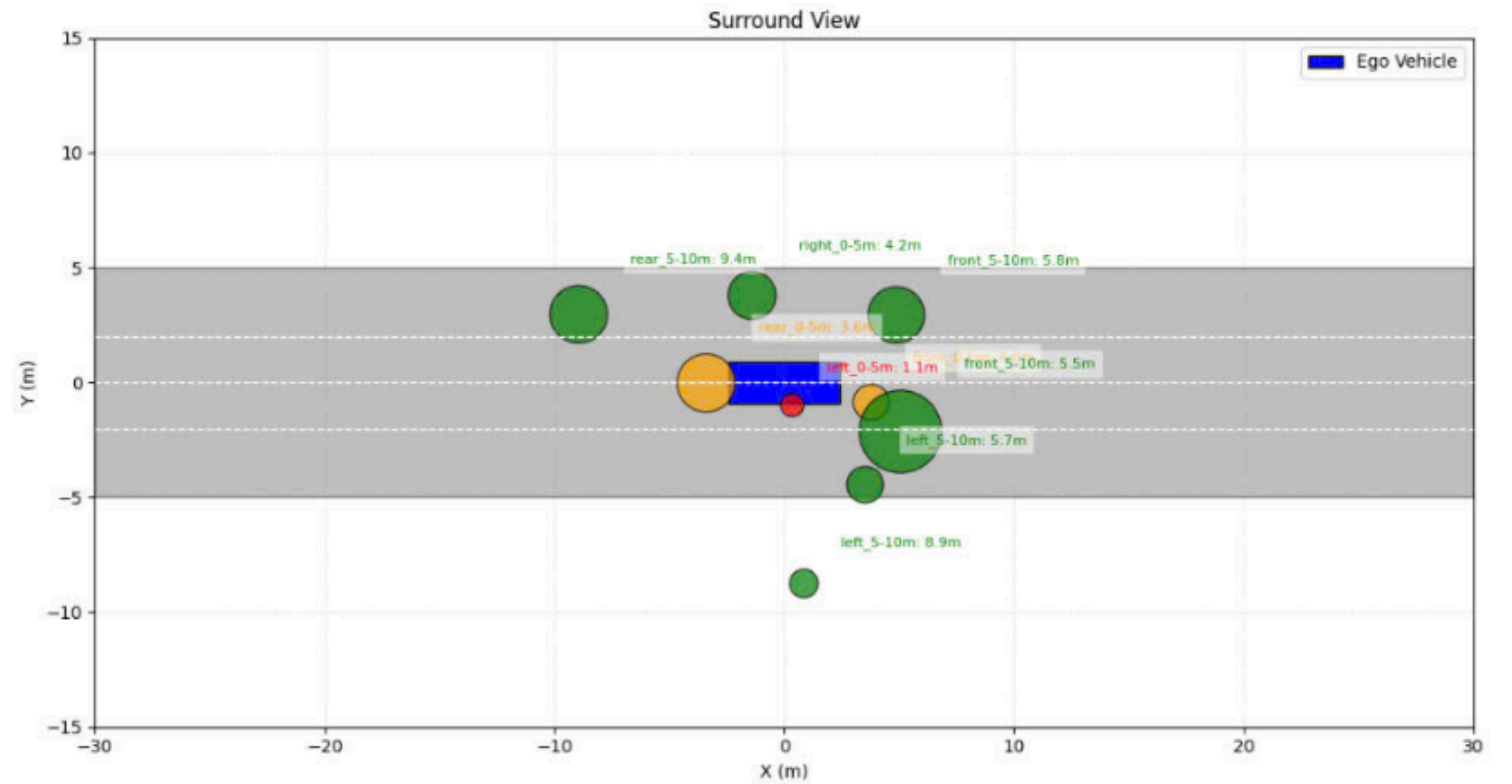
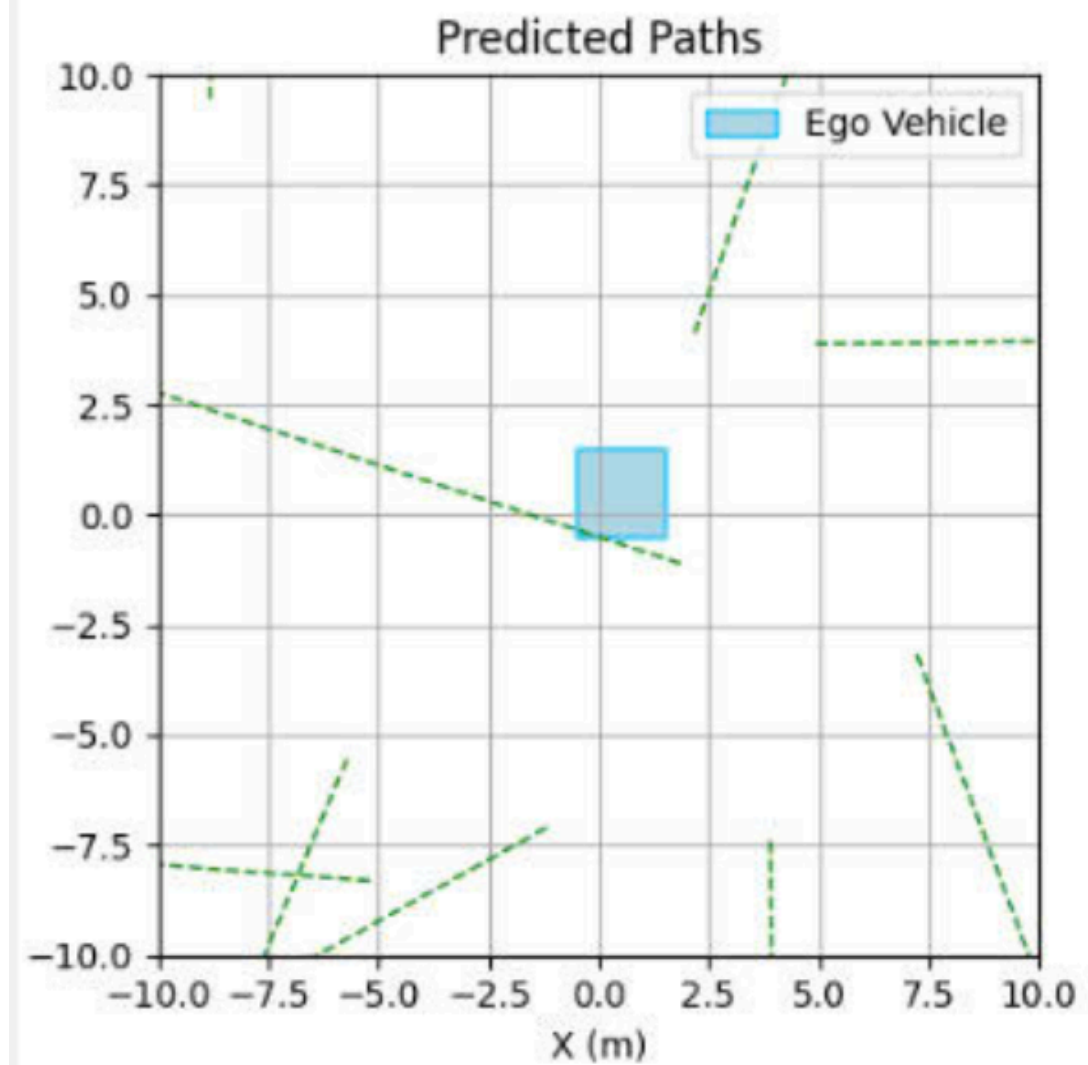


loss\_plot



prediction\_visualization

# Motion Prediction Model

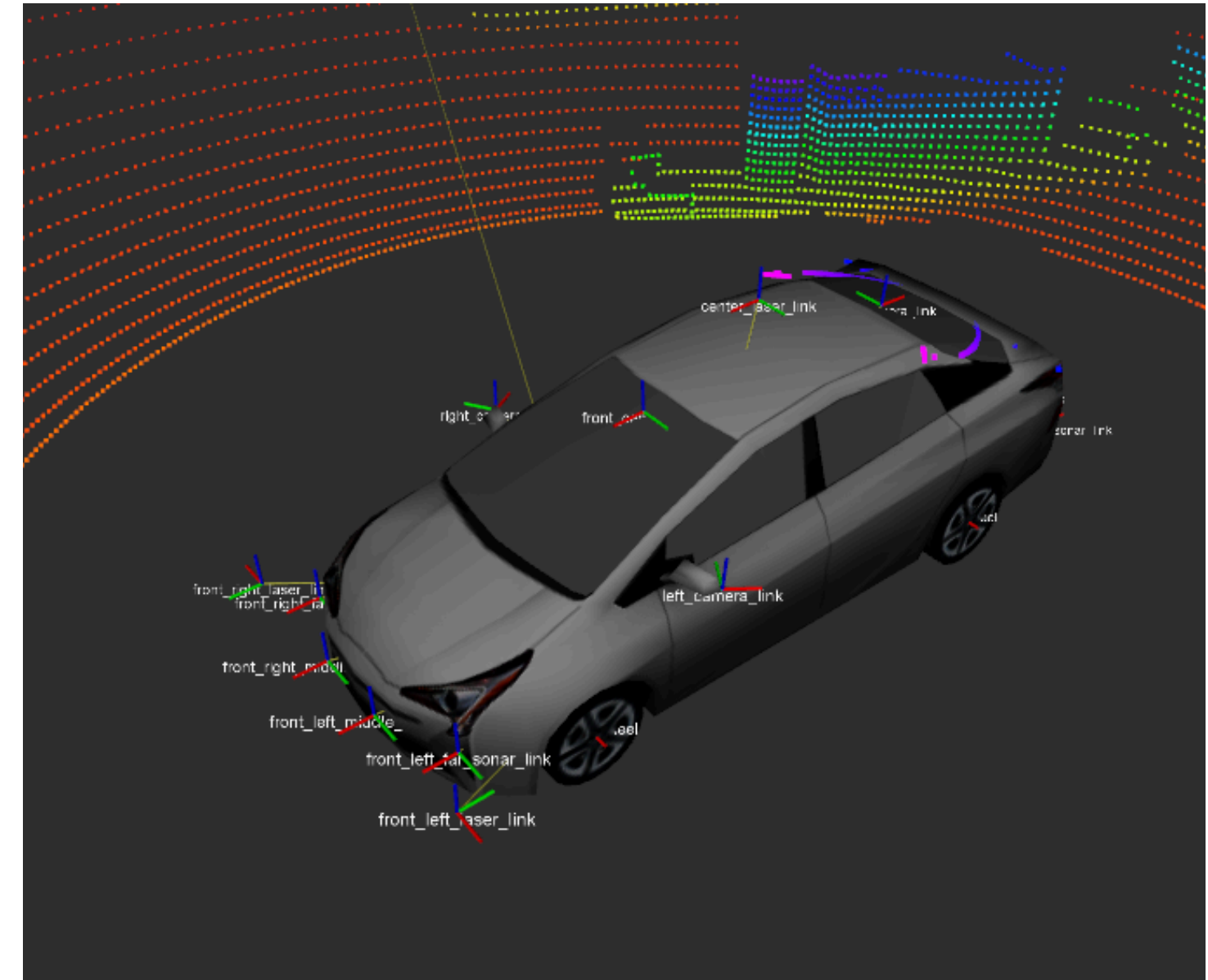




# Future plan



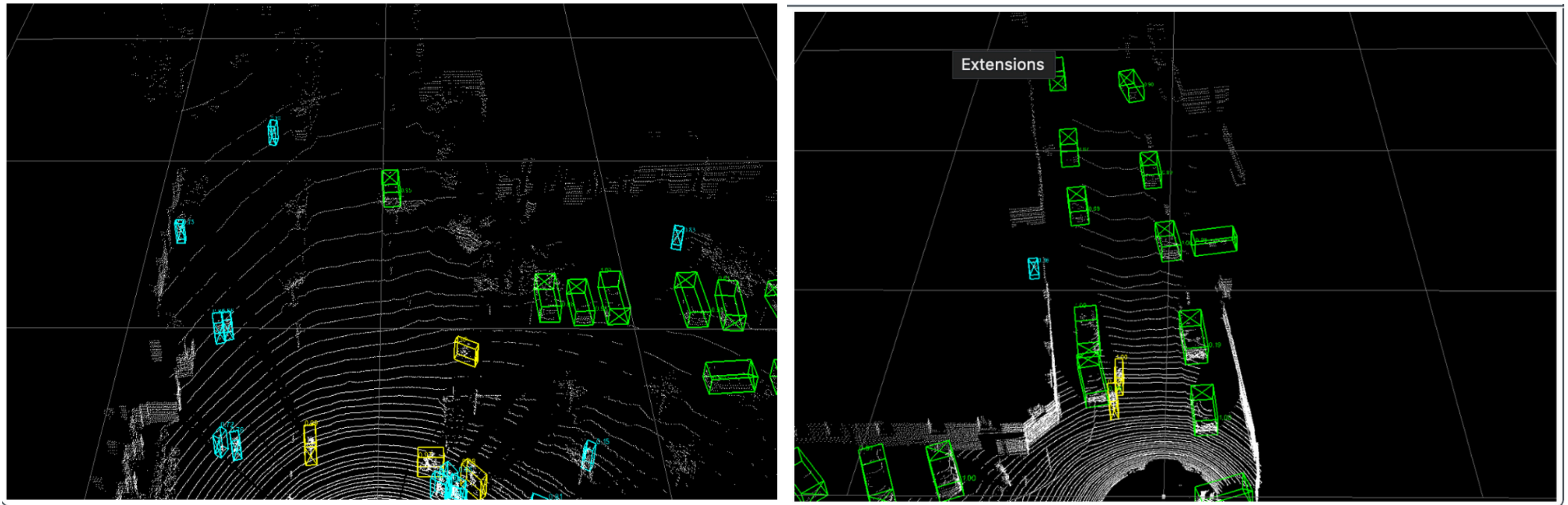
*Carla simulator*



*Gazebo simulator*

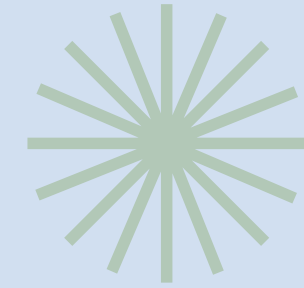
# Future plan

Improving Object Detection Granularity and Classification Accuracy



<https://github.com/open-mmlab/OpenPCDet/tree/master>

# Scalability



## From Vehicles to Smart Traffic Systems



Initially for vehicles, the system can later integrate with smart infrastructure, providing data for urban traffic management.

## Integration with Autonomous Vehicles



The system can supply critical data to autonomous driving systems, optimizing route planning and reducing collision risks.



# THANKS FOR LISTENING

